

Hospital Pathway Design Decisions and Justification

Submission: AISD (UFCEP6-0-3) - Justification of Decisions

Team: Ruby (A), Estrella (B), Ender (C), Ryan (D)

Submission baseline: Git commit `fc6293e`; executable process `Hospital_All_Processes_Simple_C8`

Formal package: `Coursework/hospital-pathway/`

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Purpose and scope

This document explains the choices behind the group's operational Camunda 8 model. The model brings referral, first visit, clinical care, letters, funding, treatment booking, changes, enquiries and reporting onto one executable canvas. It is a classroom demonstration using fictional case IDs and simulated external services. It is not a production hospital system. The editable BPMN, eleven editable Camunda Forms and Java source are in the formal package above.

The business rules were taken from the supplied hospital case study and are indexed as BR-01 to BR-42 in `docs/requirements-traceability-matrix.md`. The decisions below distinguish what the BPMN actually executes from what its annotations explain as a business policy.

Whole-group ownership and implementation map

The P-numbers are business-process references, not four separate executable processes or exclusive ownership of every shared node. The same Register, Clinical care or Funding task can serve more than one member's route. Each member must be able to explain the integrated hand-off as well as their own contribution.

A / Ruby

Main route and human forms: P5 clinical care, P7 funding/adjustment, P8 urgent authorisation, P13 report review; `clinical_care`, `funding_assessment`, `finance_adjustment`, `urgent_authorisation`, `simple_record`

Java and domain contribution: `RefundWorkers`: `request-refund` and `mark-payment-investigate`; writes `payment_ledger` and `audit_event`. Explains the shared `request-payment` message worker on the funding route.

Verification boundary: Five current Java tests pass; older group S1-S10 plan is not an execution record. P8/P13 and the final integrated refund route still require recorded acceptance runs.

B / Estrella

Main route and human forms: P3/P4 first visit and P6 treatment booking; `book_visit` and `book_treatment`

Java and domain contribution: Designed the shared H2 domain schema and `BookingWorkers`: `check-slot`, `reserve-appointment`, `flag-resource-unavailable`; writes `booking_slot` and audit events. Explains the shared `booking-confirmation` worker.

Verification boundary: Historical paths exist for an earlier booking/payment model and the original whole-hospital version; no current B-specific Java database test or final-version booking run is committed.

C / Ender

Main route and human forms: P1 request registration, P2 consultant referral decision/redirect, P9 clinic letter; register_request, review_referral, dispatch_letter, with simple_record on redirect or delay

Java and domain contribution: LetterWorkers: dispatch-clinic-letter and notify-referrer; writes keyed mock receipts to audit_event.

Verification boundary: The final-model G07 run records approved-letter dispatch and process variables. A direct audit-row assertion, redirected-referrer route and C-specific retry test are still open.

D / Ryan

Main route and human forms: P10 three enquiry outcomes and P11/P12 clinical change; simple_record and clinical_change

Java and domain contribution: CareChangeWorkers: notify-care-change; writes/reuses an audit receipt and reports whether Finance review is required.

Verification boundary: Seven current Java tests and four BPMN structural checks pass; final-version live care-change acceptance is still unrecorded.

The shared HospitalPathwayApplication deploys the one BPMN and eleven forms. HospitalPathwayWorkers contains request-payment and send-booking-confirmation; these are group integration points, not evidence that a single member implemented all process steps. The ownership allocation follows Est/est: 外部工作器与领域数据库总指引.md; the formal Java and BPMN files, rather than older learning copies, define current executable behaviour.

Decision 1 One integrated executable process

Choice. We use one executable hospital process inside a collaboration diagram. A single start event leads to Register request. The request-type gateway selects referral, existing-care change, administrative/clinical/financial enquiry or management report. The referral route can progress through a visit, clinical care, correspondence, funding and treatment. Distinct end events show rejected referral, resolved enquiry, completed care and other completed outcomes.

Reason. One process allows the group to demonstrate a complete patient pathway, shared variables and hand-offs rather than four disconnected demonstrations. The gateway makes the non-referral requests explicit without pretending that P1 to P13 are a mandatory numerical sequence.

Alternative and trade-off. Four smaller member-owned BPMN models would be easier to read and deploy independently, but would not show the integrated hand-offs in one instance. The integrated model is wide and requires a reading guide. If the large model cannot deploy in the demonstration environment, the teacher's announcement permits smaller, independent models; each such model would still need its own start/end, worker and bound forms. The current submission remains the integrated model.

Decision 2 Participants responsibilities and system boundaries

Choice. The Hospital is the executable pool, divided into Administration, Clinical team and Finance lanes. The referring organisation, patient or authorised representative, and external payment provider are black-box participants. The hospital controls its user tasks, gateways, Java workers and domain database; it does not model the internal processes of those external parties.

Reason. A consultant makes referral, treatment and clinical-change decisions. Administrative staff register requests, arrange visits, process correspondence and record contact. Finance assesses funding and approves adjustments. The lanes make these responsibilities visible and keep a clinical decision separate from a financial one. Message flows show where documents, patient contact and payment information cross an organisational boundary.

Alternative and trade-off. Separate executable pools would require additional independently deployed processes and correlation rules for every exchange. Black-box external pools keep the student model runnable while retaining the business boundary. The visual message flows for referral documents and patient contact are explanatory; only `payment-result` is a runtime BPMN message in the current implementation.

Control limitation. All current user tasks are assigned to the classroom account `demo`. Lanes and the `staffRole` form field do not enforce role-based access. PB-18 AT-03 observed that a secretary-labelled submission could accept a referral. Role enforcement remains PB-22 work and must not be claimed as delivered.

Decision 3 Human tasks and Camunda Forms

Choice. The model uses nineteen user-task occurrences bound to eleven editable Camunda Forms. Task-specific forms collect referral decision, visit outcome, clinical decision, funding status, treatment resources and clinical-change authorisation. The reusable `simple_record.form` captures a human action and reason on several supporting routes.

Reason. Human judgement remains at decisions the case assigns to authorised staff. Forms give Tasklist a visible interaction and create named process variables. Output mappings turn a selected `action` into route variables such as `referralDecision`, `fundingStatus`, `careDecision`, `changeTarget` and `moneyAffected`. FEEL conditions on outgoing sequence flows then choose the next activity.

Alternative and trade-off. A separate form for every task would give more specialised validation but multiply maintenance work. Reusing the simple record form keeps the assignment manageable; the group must still check that its general fields are suitable for each bound task. A form's required-field rule is input validation, not proof that the current user has the clinical or financial authority stated in a lane.

Decision 4 Exclusive gateways and business rules

Choice. Exclusive gateways express one outcome at a time: request type, referral decision, visit outcome, clinical plan, letter status, funding/payment result, treatment resource availability, clinical-change authority and financial impact. Default routes return missing or unrecognised decisions to a relevant human task rather than silently approving them.

Reason. These gateways make the group's business rules inspectable in the BPMN XML and demonstrable in Tasklist. For example, rejection ends without first-visit booking; a pending booking does not advance to clinical care; an unauthorised change returns for correction; and a money-affected change reaches Finance rather than issuing a refund in the care-change worker.

Alternative and trade-off. Worker-only branching would hide business decisions in Java. Gateways keep routing in the model, while Java handles simulated external work and database receipts. The trade-off is that

form keys, output mappings and FEEL expressions must stay aligned. The repository includes a structural BPMN test to detect part of that drift.

Decision 5 Runtime message interaction

Choice. The `request-payment` send task is handled by a Java worker. That worker simulates a provider result and publishes the BPMN message `payment-result` with `case_reference` as the correlation key. The intermediate catch event `PaymentReceived` waits for that message before the `payment-result` gateway proceeds.

Reason. A request and a later response are different events. The message catch demonstrates asynchronous interaction and makes the relationship between the BPMN model and the Java client concrete. A non-empty patient/case ID is required for correlation.

Alternative and trade-off. The worker could complete and return variables synchronously without a message. That would be simpler but would not demonstrate message interaction. The current provider result is simulated by the worker, not returned by a live payment service. Other diagrammatic message flows are not separate runtime handlers.

Decision 6 External workers and task allocation

Choice. Ten BPMN service/send task types have matching `@JobWorker` methods. Shared workers handle `request-payment` and `send-booking-confirmation`. Member A covers refund and payment investigation; B covers slot checking, treatment reservation and resource-unavailable recording; C covers clinic-letter dispatch and referrer notification; D covers care-change notification. `HospitalPathwayApplication` deploys the BPMN and forms, and Spring Boot keeps the workers active. The hand-off is visible in each part of the model: A follows `Funding` and `FinanceAdjustment`, B follows `BookVisit` and `BookTreatment`, C follows `DispatchLetter` and `Redirect`, and D follows `ClinicalChange`.

Reason. A human completes the form, Camunda creates the matching job, Java reads process variables, performs the simulated action, returns result variables and lets the token continue. This is visible in the model and can be explained by each member. H2-backed workers use stable keys for important retry cases instead of blindly adding a new refund or notification receipt.

Alternative and trade-off. Implementing real scheduling, mail and payment integrations would add credentials, network dependence and clinical-data risk beyond the course demonstration. The mock workers make the process repeatable. They do not prove actual delivery of a letter, payment capture by a provider or booking in a hospital system.

Decision 7 Separate domain database and audit evidence

Choice. The Java application uses a file-based H2 database at `./data/hospital-domain` when launched from `Coursework/hospital-pathway/`. It contains `payment_ledger`, `booking_slot` and `audit_event`. The folder is ignored by Git and is separate from the Camunda/c8run engine database. Tests use in-memory H2 so they do not alter local demo records.

Reason. A domain database lets the group show receipts, booking occupancy and audit records after a worker runs. Case references and idempotency keys support inspection and basic retry protection. The repository commits schema and source code, not local patient data or database files.

Trade-off and limitation. A file database is sufficient for one local student demonstration, not a multi-user production service. Current successful request-payment is a simulated message interaction; a successful payment ledger row is not guaranteed by that worker. Acceptance criteria must check the variables and message, not invent a ledger row.

Decision 8 Exceptions safety and known limits

Choice. The executable model includes explicit routes for missing referral information, rejection/redirect, pending visit or treatment resources, payment failure or missing confirmation, urgent clinical authorisation, delayed letters, care changes and enquiries. A missing payment confirmation goes to investigation rather than directly to treatment booking. A clinical change with financial impact is sent to Finance for a separate human decision.

Reason. Exception paths keep an unfinished case visible to a responsible human and avoid silently treating a failed external action as success. The urgent-care route records clinician authorisation before the payment exception is allowed.

Limitations. The classroom model has no production role-based access, no real timers or escalation scheduler, no live external scheduling/payment/correspondence service and no guarantee of immutable audit storage. Source review also found that the B booking worker requests CONFIRMED for first-visit slot checking without testing a pending visit and does not promote an existing PENDING treatment record to CONFIRMED on a later ready result. This remains a code-level finding until a database test or an appropriate pending-to-ready engine run confirms its effect. The final-model G07 self-pay route completed locally, but it does not test these exceptions, browser-side form validation or every member-specific branch. These limits are stated in the accompanying acceptance results rather than counted as passes.

Verification and review boundary

At source commit `fc6293e`, a clean Maven test run passed twelve Java tests (five A refund/investigation tests and seven D care-change/database tests). `python scripts/test_member_d_bpmn.py` passed four structural checks across the model references and worker types. A new local Camunda 8 REST run of the same final BPMN source (SHA-256 recorded in [Coursework/hospital-pathway/evidence/Group_Final_E2E_20260929-135729.json](#)) completed nine user tasks and reached COMPLETED with no active incident; payment-message, booking and letter worker logs are preserved. It proves that one selected self-pay path operates across A/B/C hand-offs, not D's separate route. B's pending/ready database transition, C's referrer/retry path and D's live change route still need their own checks. The existing PB-14 and PB-18 evidence uses an earlier deployed process version 6 and proves only its listed checks. The group acceptance document records these boundaries and the failed role-permission test.

Primary repository sources: [Coursework/hospital-pathway/bpmn/W02_Hospital_All_Processes_Clean_Lines_Camunda8.bpmn](#); [Coursework/hospital-pathway/forms/](#); [Coursework/hospital-pathway/src/main/java/](#); [Coursework/hospital-pathway/src/main/resources/application.yaml](#); [docs/requirements-traceability-matrix.md](#); [Est/est: 领域数据库设计.md](#); [Est/est: 全院流程路径记录.md](#); [Ender/2026-09-28/C_外部工作器与数据库说明.md](#); [Ruby/hospital-pathway-zh-learn/验收测试计划-S1至S10-中英对照.md](#); [Ryan/2026-09-26/PB-14_acceptance-test-plan_2026-09-29.md](#); [Ryan/2026-09-26/PB-18_acceptance-test-plan_2026-09-29.md](#).